

Ali Khoramfar

Department of Electrical & Computer Engineering, University of Tehran, Iran

✉ Alikhorramfar@gmail.com in LinkedIn GitHub Google Scholar

Research Interests

- Vision-Language Models • Multilingualism and Cross-Lingual NLP • Clinical and Biomedical Applications

Education

University of Tehran

2023 – Present

Master of Science in Computer Engineering

- **GPA: 19.33/20** (4/4)
- **Relevant Courses:** Natural Language Processing, Deep Learning, Machine Learning, Social Networks, Statistical Inference, Distributed Machine Learning
- **Master's Thesis** (Supervisors: Prof. Hesham Faili & Prof. Mohammad Javad Dousti):
 - **Title:** Investigating the Performance of Multimodal Large Language Models for Medical Image Analysis
 - **Summary:** Analyzing the performance of multimodal large language models on complex reasoning tasks and multilingual settings, with medical image analysis employed as a representative application.

JundiShapur University of Technology

2017 – 2022

Bachelor of Science in Computer Engineering

- **GPA: 19.11/20** (4/4)
- **Ranked 1st** among all Computer Engineering students
- **Relevant Courses:** Artificial Intelligence and Expert Systems, Image Processing, Internet Engineering, Microprocessors, Principles of Database Design
- **Bachelor's Project** (Supervisor: Dr. Maryam Chinipardaz):
 - **Title:** Green Internet of Things and Solar Energy
 - **Summary:** Designed a five-layer architecture to optimize IoT energy efficiency and investigated solar-powered solutions for sustainable deployment.

Publications

Conference Papers

- *Stable Evidence, Unstable Decisions: An Empirical Analysis of Model Decision Stability in VLMs*
Khoramfar A., Dousti M. J., Mohamadian A., & Faili H. **ACL 2026 (Findings)**
- *DeepQuestion: Systematic Generation of Real-World Challenges for Evaluating LLMs Performance*
Khoramfar A., Ramezani A., Mohajeri M. M., Dousti M. J., Nili Ahmadabadi M., & Faili H. **LREC 2026**
- *PerMed-MM: A Multimodal, Multi-Specialty Persian Medical Benchmark for Evaluating Vision Language Models*
Khoramfar A., Dousti M. J., & Faili H. **IJCNLP-AAACL 2025 (Main Conference)**

Journal Papers

- *Green Internet of Things and Solar Energy*
Chinipardaz, M., Khoramfar, A., & Amraee, S. (2024). Environmental Science and Pollution Research, 31(12), 18296–18312. [\[DOI\]](#)

In Preparation

- *From Machine Learning to VLMs: A Comparative Study for MRI-Based Brain Tumor Differentiation*
Khoramfar A., Mohamadian A., & Dousti M. J.

Other Publications

- *Network Level Energy Solutions for Green Internet of Things*
Khoramfar A., & Chinipardaz M. (2023). First National Conference on Internet of Things
- *Green Computing Solutions in Environmental Protection and Use of Renewable Energy*
Amraei S., & Khoramfar A. (2021). National Conference on Chemical Engineering & Nanotechnology (in Persian)
- *The Role of IoT in Innovation and Business Development*
Khoramfar A., & Chinipardaz M. (2021). First National Conference on Entrepreneurial Universities (in Persian)

Research Experiences

Graduate Student Researcher, University of Tehran <i>Natural Language Processing Laboratory</i> - Conducted research on multimodal and multilingual evaluation of LLMs and VLMs, focusing on reasoning, benchmark construction, and low-resource languages. - Led dataset curation and large-scale experimental evaluations, contributing to multiple research papers. - Reviewed submissions for ACL Rolling Review, providing detailed technical and methodological feedback.	<i>2023 – Present</i>
Undergraduate Research Assistant, JundiShapur University <i>Computer Networks Laboratory</i> - Engaged in research on energy-efficient computing. - Conducted literature reviews and collaborated on writing and revising research manuscripts.	<i>2020 – 2022</i>

Teaching Experiences

Distributed Machine Learning – TA <i>Supervisor: Prof. Mohammad Javad Dousti</i> Designed assignments focusing on scalability, load balancing, and communication–computation trade-offs in distributed machine learning systems.	<i>Sep 2025 – Feb 2026</i>
Essential Computing Skills – Head TA <i>Supervisor: Prof. Mohammad Javad Dousti</i> - Coordinated course logistics, including schedule management and preparation of course materials. - Assisted with grading and facilitated technical workshops on Vim, Git, Docker and Kubernetes.	<i>Jan 2025 – Sep 2025</i>
Natural Language Processing – TA <i>Supervisor: Prof. Heshaam Faili</i> Assisted in assessment design and grading and developed a project on Vision-Language Models (CLIP), introducing students to multimodal embeddings, similarity analysis and bias evaluation in pre-trained models.	<i>Aug 2024 – Feb 2025</i>
Deep Learning – TA <i>Supervisor: Dr. Ahmad Kalhor</i> Designed and supervised a course project introducing students to Vision Transformers, including data pre-processing, transfer learning, fine-tuning strategies and comparative evaluation with CNN architectures.	<i>Aug 2024 – Feb 2025</i>
Internet Engineering – TA <i>Supervisor: Dr. Maryam Chinipardaz</i> Delivered technical workshops on web hosting and search engines, covering hosting configuration, domain management, deployment, indexing, ranking algorithms and SEO optimization practices.	<i>Sep 2021 – Jan 2022</i>

Academic Projects

Distributed Machine Learning [GitHub] <i>Supervisor: Prof. Mohammad Javad Dousti</i>	<i>Sep 2024 – Jan 2025</i>
Natural Language Processing [GitHub] <i>Supervisor: Prof. Heshaam Faili</i>	<i>Jan 2024 – Jul 2024</i>
Deep Learning [GitHub] <i>Supervisor: Dr. Ahmad Kalhor</i>	<i>Jan 2024 – Jul 2024</i>
Machine Learning [GitHub] <i>Supervisor: Dr. Tavassolipour</i>	<i>Jan 2024 – Jul 2024</i>
Statistical Inference [GitHub] <i>Supervisor: Dr. A. Dehaqani</i>	<i>Sep 2023 – Jan 2024</i>

Technical Skills

Programming: Python (NumPy, Pandas, Matplotlib, scikit-learn, PySpark), C++, C#, SQL
Frameworks: PyTorch Distributed, Hugging Face (Transformers, Datasets, PEFT), LangGraph, Unsloth
Tools: Git, MPI, Slurm

Languages

English: Professional Working Proficiency
Persian: Native